

QUESTION PAPER - SET 5

SECTION 1 - PERSONAL DATA

1. Name

2. Mobile Number

3. Class 10 Overall Percentage

4. Class 12 PCM Percentage

5. What is your percentage in Pre Sea Training till 4th Semester?

6. Tell us about your strengths.

7. Tell us about your weaknesses.

8. Have you participated in any extracurricular activities during your Pre Sea Training? If

9. yes, share details.

10. Tell us about your family.

11. Where do you see yourself after 5 years?

SECTION 2 - COMPREHENSION PASSAGE

In the late 21st century, the Global Council enacted the "Resource Equilibrium Accord" to address escalating conflicts over dwindling freshwater reserves. The Accord introduced a tiered quota system, assigning water access rights based on each nation's ecological footprint, technological capability for desalination, and per-capita consumption history.

While wealthier nations invested in advanced desalination plants and atmospheric water harvesters, poorer nations — despite having smaller ecological footprints — struggled to meet their citizens' basic needs under their reduced quotas.

Critics argued that the Accord unintentionally entrenched inequality, as historical consumption patterns favored nations that had previously overused water.

To address the backlash, an amendment proposed the creation of a "Water Equity Fund" financed by a surcharge on nations exceeding their quota. The fund would support sustainable water projects in under-resourced regions.

Yet, debates continued over enforcement: how to monitor usage accurately, prevent data manipulation, and ensure that the fund genuinely benefited those in need.

1. What was the main goal behind the introduction of the Resource Equilibrium Accord in the late 21st century?

- A. To establish a global water trading market
- B. To balance global water distribution and prevent conflicts over freshwater resources
- C. To reward nations with low per-capita water consumption
- D. To increase funding for desalination research

2. Which aspect most contributed to the Resource Equilibrium Accord unintentionally reinforcing inequality among nations?

- A. Technological advancements in desalination
- B. Inaccurate monitoring of water usage
- C. Historical consumption patterns that benefited countries with prior high water use
- D. Opposition to the Water Equity Fund

3. Which conclusion is best supported by the information provided in the passage about the Resource Equilibrium Accord?

- A. The amendment eliminated all concerns about fairness
- B. Nations with smaller ecological footprints were given larger quotas
- C. Wealthier nations opposed atmospheric water harvesting
- D. The Accord unintentionally reinforced pre-existing inequalities

4. Under the Resource Equilibrium Accord, when would a nation be required to pay a surcharge to support the Water Equity Fund?

- A. When its per-capita consumption is below the global average

- B. When it fails to reduce its ecological footprint
- C. When it does not participate in atmospheric water harvesting
- D. When its water usage exceeds the assigned quota

5. What is the main challenge that remains unresolved regarding the amendment to the Resource Equilibrium Accord?

- A. Lack of sustainable water project ideas
- B. The unwillingness of wealthier nations to pay surcharges
- C. Uncertainty in the fund's revenue source
- D. Ensuring transparency and preventing misuse of funds

SECTION 3 - COMPREHENSION PASSAGE

The Sundarbans, located in the delta region of India and Bangladesh, is the largest mangrove forest in the world. It is home to diverse wildlife, including the famous Royal Bengal Tiger. The mangroves act as a natural barrier, protecting coastal areas from cyclones and storm surges. They also provide livelihoods for local communities through fishing, honey collection, and eco-tourism. However, climate change poses a serious threat to the Sundarbans. Rising sea levels, increasing salinity, and frequent storms are causing habitat loss.

Conservation projects are being implemented, focusing on planting more mangroves, reducing illegal logging, and promoting sustainable fishing practices. International cooperation is considered vital for the long-term preservation of this unique ecosystem.

1. Which of the following is NOT listed as a livelihood activity for local communities in the Sundarbans, according to the passage? (1 Point) Select the option that is not mentioned as a source of livelihood.

- A. Agriculture
- B. Honey collection
- C. Fishing
- D. Eco-tourism

2. Which of the following is a direct impact of climate change on the Sundarbans ecosystem? (1 Point) Select the option that best represents a climate change effect described in the passage.

- A. Increasing salinity of water
- B. Introduction of new wildlife species
- C. Growth in honey collection
- D. Expansion of fishing zones

3. What is one of the key protective roles played by the mangrove forests in the Sundarbans? (1 Point) Select the option that best describes how mangroves benefit the coastal environment.

- A. They increase the frequency of tidal waves.
- B. They are a major source of underground minerals.
- C. They supply drinking water to nearby villages.
- D. They act as a shield against cyclones and storm surges.

4. What is the primary objective of the conservation efforts described in the Sundarbans passage? (1 Point) Select the option that best reflects the main goal of these projects.

- A. To increase honey production for export
- B. To encourage local communities to move to urban areas
- C. To restore and safeguard the mangrove forests and their biodiversity
- D. To expand commercial fishing operations

5. Why does the passage emphasize the necessity of international cooperation for preserving the Sundarbans? (1 Point) Select the option that best explains the importance of cross-border collaboration.

- A. The Sundarbans ecosystem spans across India and Bangladesh, requiring joint efforts.
- B. Local communities are unwilling to support conservation projects.
- C. The region attracts international tourists who fund conservation.
- D. Climate change is only a concern for a few developed countries.

SECTION 4 - ENGLISH APTITUDE

1. Choose the correct option to complete the sentence: "You should not leave the meeting _____ you have clarified all your doubts."

- A. Even if
- B. Unless
- C. Provided
- D. Despite

2. Choose the word that is most nearly opposite in meaning to "lucid".

- A. Clear
- B. Vague
- C. Logical
- D. Coherent

3. Select the word that best completes the sentence: "The technical report was so _____ that even seasoned professionals struggled to understand it."

- A. Obscure
- B. Redundant
- C. Ambiguous
- D. Verbose

4. Select the sentence that has the same meaning as: "The cadet received a warning for not adhering to procedures."

- A. The cadet received a warning because he did not adhere to procedures.
- B. The cadet received a warning although he adhered to procedures.
- C. The cadet received a warning so that he would adhere to procedures.
- D. The cadet received a warning even though he adhered to procedures.

SECTION 5 - NUMERICAL APTITUDE

1. A cube has opposite faces marked as: (1, 6) (2, 5) (3, 4) If face 1 is on the top and face 2 is on the front, which number is on the right face?

- A. 3
- B. 4
- C. 5
- D. 6

2. Two boats are anchored on opposite sides of a 100 m tall lighthouse. From the boats, the angles of elevation to the top of the lighthouse are 30° and 45° respectively. What is the distance between the two boats?

- A. 273 m
- B. 173 m
- C. 200 m
- D. 300 m

3. The length of a rectangle is twice its breadth. If its area is 200 square units, what is the breadth?

- A. 5
- B. 8
- C. 10
- D. 20

4. Two workers can complete a job in 10 hours and 15 hours respectively. If they work together, how long will it take them to finish the job?

- A. 6 hours
- B. 9 hours
- C. 12 hours
- D. 7 hours

5. Which set of figures best illustrates the transformation where closed shapes gradually open up and open shapes gradually close? [IMAGE QUESTION]

- A. Option 1
- B. Option 2
- C. Option 3
- D. Option 4

SECTION 6 - MARITIME KNOWLEDGE

1. Which frontal type is MOST commonly associated with the process shown in the diagram?

- A. Stationary front
- B. Warm front
- C. Cold front
- D. Occluded front

2. In ship stability, GM refers to:

- A. Gross Measurement
- B. General Metric
- C. Metacentric Height
- D. Gravity Measurement

3. On a nautical chart, which color is commonly used to indicate shallow water and the boundaries of certain regulated areas?

- A. Gold
- B. Blue
- C. Black
- D. Green

4. According to ship safety regulations, which of the following must be made weathertight as part of the Conditions of Assignment?

- A. Hatchways situated on the freeboard deck
- B. Sounding pipes on deck
- C. Cargo hold ventilators below deck
- D. Scuppers above deck

5. Which part of a ship experiences the greatest bending moment in heavy seas?

- A. Midship section
- B. Forecastle
- C. Stern peak
- D. Bridge deck

6. Which part of the ship is located at the front?

- A. Stern
- B. Bow
- C. Port
- D. Starboard

7. What is the primary purpose of a marine sextant?

- A. Measuring wind direction
- B. Determining celestial altitude
- C. Measuring sea depth
- D. Determining engine power

8. What does SOLAS stand for?

- A. Safety of Life at Sea
- B. Security of Life and Ships
- C. Standard Operations for Large Ships
- D. Safety Operations and Life Saving

9. In celestial navigation, what is the First Point of Aries () primarily used to measure?

- A. Measuring True Altitude
- B. Measuring Azimuth
- C. Measuring Right Ascension
- D. Measuring Longitude

10. When crossing a traffic separation scheme, the vessel should:

- A. Cross at any convenient angle
- B. Cross at right angles to the traffic flow

- C. Alter course to avoid lanes
- D. Follow the traffic flow

11. Which celestial body is commonly used for obtaining a compass error by azimuth during twilight?

- A. Polaris
- B. Venus
- C. The Sun
- D. Mars

12. What does the term "angle of loll" describe in ship stability?

- A. The angle at which the center of gravity is vertically above the center of buoyancy
- B. The time taken for a vessel to complete one roll cycle
- C. KM is greater than GM
- D. The heel angle at which a vessel with negative GM achieves equilibrium due to hull shape

13. Which safety equipment is used to prevent drowning during emergencies?

- A. Compass
- B. Lifebuoy
- C. Sextant
- D. Radar reflector

14. At which location on Earth does the Coriolis effect reach its greatest strength?

- A. At the Equator
- B. At the Tropics
- C. At 30° latitude
- D. At the Poles

15. In celestial navigation, what is the reason for drawing the intercept at right angles to the azimuth direction?

- A. Because the azimuth determines the observer's latitude
- B. Because the Line of Position (LOP) is always perpendicular to the azimuth to the celestial body
- C. Because the intercept is affected by time zones
- D. Because the Earth's magnetic field influences the plotting

16. A vessel departs Felixstowe, England at 1000 hrs local time (GMT+0) on 2nd March 2014, bound for Kolkata, India, a distance of 10,080 nautical miles. If the vessel maintains a constant speed of 10 knots, what will be the vessel's estimated time of arrival (ETA) at Kolkata in local time (GMT+5.5)?

- A. 13th April 2014, 1000 hrs LT
- B. 13th April 2014, 0430 hrs LT
- C. 13th April 2014, 1530 hrs LT
- D. 13th April 2014, 1200 hrs LT

17. Which cloud type is most commonly associated with thunderstorms and heavy rainfall?

- A. Cirrus
- B. Stratus
- C. Cumulonimbus
- D. Altocumulus

18. How should a newly issued Preliminary Notice to Mariners (P-Notice) be marked on a nautical chart?

- A. In red marker
- B. Lightly in pencil
- C. In permanent ink
- D. Not marked until confirmed

19. In the Northern Hemisphere, what is the typical direction of wind circulation around a low-pressure area?

- A. Anticlockwise and inward

- B. Clockwise and inward
- C. Clockwise and outward
- D. Anticlockwise and outward

20. The vertical distance between the upper edge of the deck line and the centre of the load line disc represents the:

- A. Summer Freeboard
- B. Summer Draft
- C. Fresh Water Allowance
- D. Winter North Atlantic Freeboard

21. What is the purpose of a ship's whistle or horn?

- A. Entertainment
- B. Distress communication only
- C. Navigational sound signaling
- D. Engine warning

22. When approaching a danger marked by a Cardinal buoy, the vessel should:

- A. Pass on the port side
- B. Pass on the starboard side
- C. Keep to the side named by the buoy
- D. Keep the buoy on the vessel's port side

23. Aneroid barometers are typically calibrated to display atmospheric pressure in which of the following units?

- A. Inches of mercury and millibars
- B. Feet of mercury and millibars
- C. Inches of mercury and millimeters
- D. Inches of mercury and centimeters

24. Which instrument is specifically used to measure the apparent wind direction and speed on a moving ship?

- A. Barometer
- B. Psychrometer
- C. Hygrometer
- D. Anemometer

25. Ventilators forming part of Conditions of Assignment must:

- A. Be of any height if fitted with wire mesh
- B. Have minimum height and efficient closing appliances
- C. Have closing devices only above superstructure decks
- D. Be permanently open for machinery ventilation

26. In marine meteorology, what does a rapidly falling barometer usually indicate?

- A. Stable weather conditions
- B. Approaching bad weather or storm
- C. Extremely dry air
- D. Calm sea conditions

27. Which factor causes the settling error of a gyrocompass to become more pronounced?

- A. Operating at higher latitudes
- B. Operating at lower latitudes
- C. Increasing the vessel's speed
- D. Navigating in areas of high magnetic variation

28. The total area beneath the GZ (righting arm) curve from 0° to 40° of heel is a measure of which of the following vessel characteristics?

- A. The vessel's dynamic stability within that heel range
- B. The vessel's metacentric height (GM)
- C. The vessel's KM
- D. The vessel's maximum righting lever

29. Which type of weather front is most commonly linked to severe thunderstorms, heavy rain, and sudden gusty winds?

- A. Cold front
- B. Occluded front
- C. Warm front
- D. Stationary front

30. What is the SI unit used to measure atmospheric pressure?

- A. Knot
- B. Pascal
- C. Joule
- D. Newton